

Mathematics

Stage 8

Paper 1

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Draw a ring around the part of a circle that is **not** a straight line.

diameter chord radius circumference tangent

[1]

- 2 Work out.

$$-6 \times -13$$

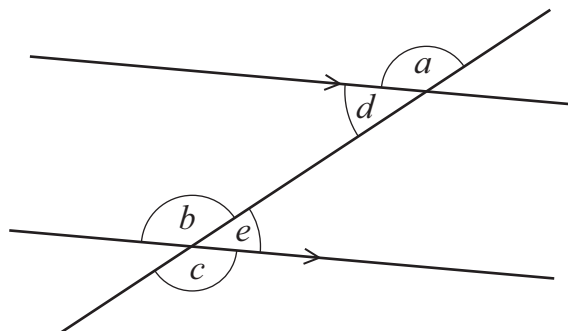
..... [1]

- 3 Draw a ring around the letter that is the subject of the formula $\pi dh + 2\pi r^2 = A$.

d h r A

[1]

- 4 The diagram shows a pair of parallel lines with a straight line crossing them and some angles marked with letters.



Complete these sentences with the correct letters.

Angles and are alternate angles.

Angles and are corresponding angles.

Angles and are vertically opposite angles.

[2]

- 5 Complete each of these calculations with the correct whole number.

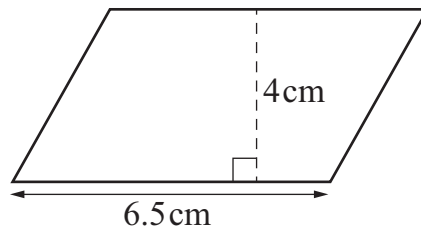
$$13^0 = \dots\dots\dots$$

$$(2^3)^2 = \dots\dots\dots$$

$$11^{15} \div 11^{13} = \dots\dots\dots$$

[3]

- 6 The diagram shows a parallelogram.



NOT TO
SCALE

Work out the area of the parallelogram.

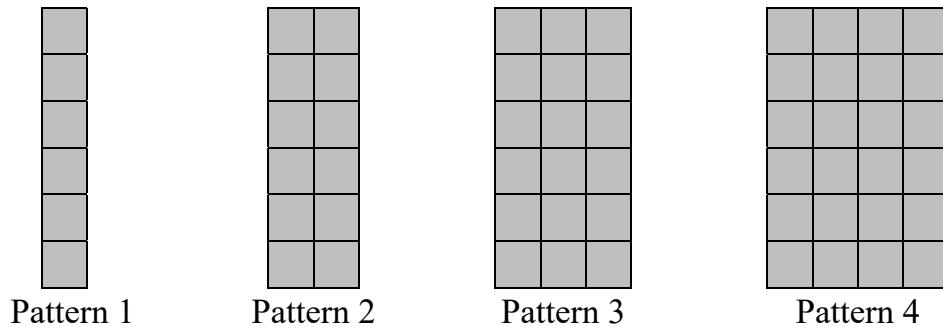
..... cm^2 [1]

- 7 Naomi draws a regular polygon with exactly 13 lines of symmetry.

Write down the order of rotational symmetry of her polygon.

..... [1]

- 8 The diagram shows the first four patterns in a sequence made with square tiles.



Draw a ring around the exact number of tiles that could make a pattern in this sequence.

32 tiles

64 tiles

96 tiles

602 tiles

[1]

- 9 Mike and Carlos share some money in the ratio 2 : 5

Draw a ring around the correct statement.

Mike gets $\frac{2}{5}$ of the money.

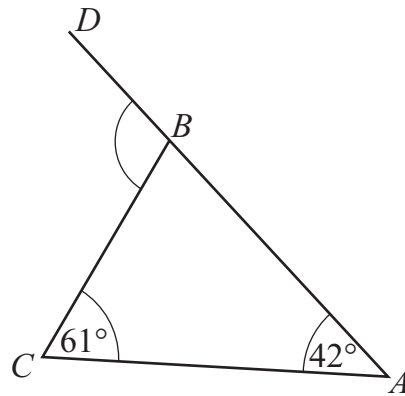
Mike gets $\frac{5}{2}$ of the money.

Mike gets $\frac{2}{7}$ of the money.

Mike gets $\frac{2}{3}$ of the money.

[1]

- 10 The diagram shows a triangle ABC .



NOT TO
SCALE

ABD is a straight line.

Find angle CBD .

.....[°] [1]

- 11 (a) The n th term of a sequence is $\frac{n}{2} + 5$

Work out the 6th term of the sequence.

..... [1]

- (b) Here are the first five terms of a different sequence.

2, 8, 14, 20, 26

Find the n th term of this sequence.

..... [2]

12 Here are the names of some quadrilaterals.

squares

parallelograms

rhombuses

kites

Choose the correct word from the list to complete this statement.

All rectangles are

[1]

13 (a) Oliver writes this calculation.

$$1 + 2 \times 5^2 = 75$$

Tick (✓) to show if Oliver is correct or **not** correct.

Oliver is correct

☐

Oliver is not correct

☐

Explain your answer.

.....

.....

[1]

(b) Work out.

$$\sqrt{3^2 \times 2^3 + 28}$$

..... [2]

- 14 Draw a bearing of 285° from point X .



[1]

- 15 Work out.

$$5\frac{7}{12} - 3\frac{3}{4}$$

Give your answer as a mixed number in its simplest form.

..... [3]

- 16** M is the point $(1, 3)$.
 M is the midpoint of the line AB .

Complete the coordinates for A and B .

$$A = (\quad -2 \quad , \quad \text{.....} \quad)$$

$$B = (\quad \text{.....} \quad , \quad \quad 7 \quad)$$

[2]

- 17** Here is a mathematical statement.

$$56 \text{ kilometres} + 4.3 \text{ miles} = x \text{ miles}$$

Work out the value of x .

$$x = \quad \text{.....} \quad [2]$$

18 Find the value of each expression when $x = -2$ and $y = 5$

$$4(y - x) = \dots\dots\dots$$

$$10y - 3x^2 = \dots\dots\dots$$

[2]

19 Jamila has a 6-digit number.
Two of the digits in her number are hidden.

4 ■ ■ 3 5 6

Tick (✓) to show if each of Jamila's statements must be true, could be true or must be false.

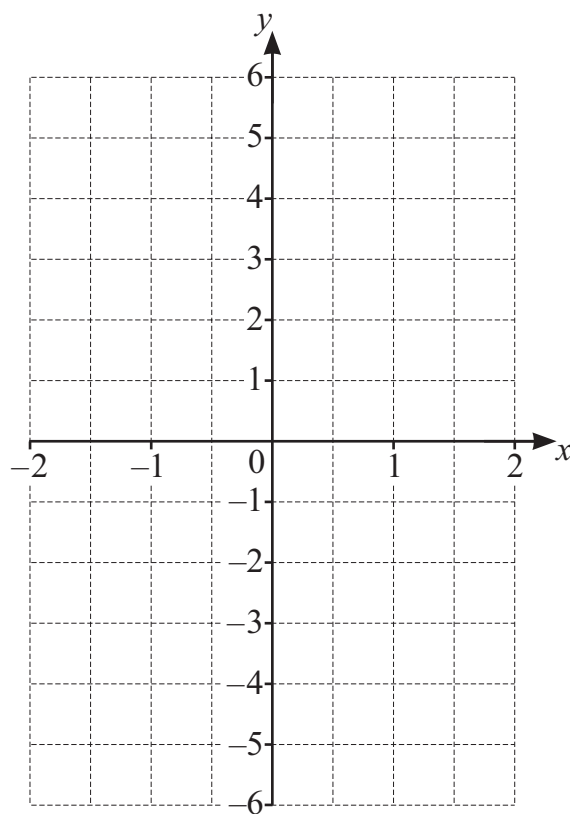
	Must be true	Could be true	Must be false
My number is divisible by 4			
My number is divisible by 5			
My number is divisible by 8			
My number is divisible by 9			

[2]

20 Complete the table for $y = 2x - 1$

x	-2		2
y		-3	

Draw the graph of $y = 2x - 1$ on the grid.



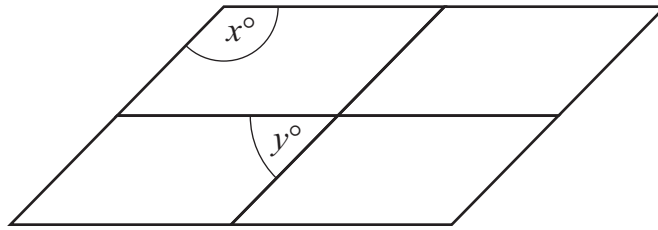
[3]

21 Eva says, ‘The highest common factor of 36 and 60 is 6’

Show that she is **not** correct.

[1]

- 22 The diagram shows a tessellation of four parallelograms with no lines of symmetry.



Write down a possible pair of values for x and y .

$x =$

$y =$

[1]

- 23 Factorise.

$$12x^2 + 18x - 30xy$$

..... [2]

- 24 Complete each box with the correct power.

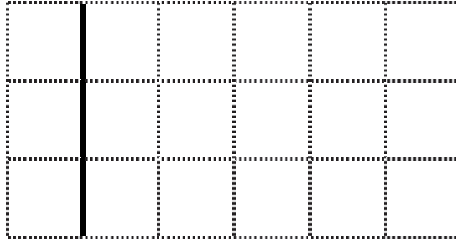
$$48 \times 150 = 2^{\square} \times 3^{\square} \times 5^{\square}$$

[2]

25 These are the lengths, in centimetres, for a sample of 10 freshwater fish.

6.9 7.6 7.8 6 5.7 6.5 5.1 7.2 7.1 7.8

(a) Complete the stem-and-leaf diagram for this information.



Key:

[3]

(b) The table shows information about the lengths of 10 **saltwater** fish in a different sample.

	Saltwater
Median	8.4 cm
Range	5.8 cm

Write **two** comparisons between the distribution of the lengths of the freshwater fish and the distribution of the lengths of the saltwater fish.

Give your comparisons in context.

You **must** include the statistics you use.

1

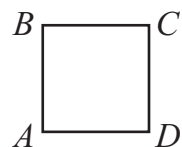
.....

2

.....

[4]

26 Square $ABCD$ has a side length of 5 units.



$ABCD$ is drawn on a grid so that C is at $(16, 11)$.

$ABCD$ is then translated 4 left and 2 down.

Find the coordinates of vertices A and B after this translation.

$A = (\dots\dots\dots, \dots\dots\dots)$

$B = (\dots\dots\dots, \dots\dots\dots)$

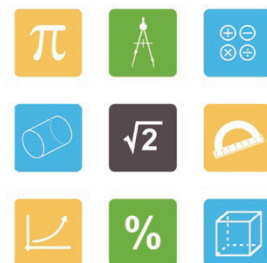
[2]

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Mathematics

Stage 8

Paper 1

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Mark scheme annotations and abbreviations**

FT	follow through after error
dep	dependent
oe	or equivalent
cao	correct answer only
isw	ignore subsequent working
soi	seen or implied

Difference in printing

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square brackets in mark scheme

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

The table

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

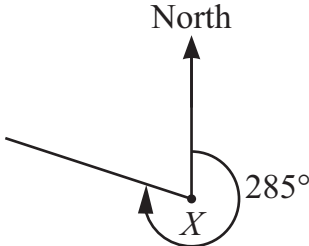
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer The graph line passing through a point implies the point even though there is no cross	A horizontal line and vertical line from the axes meeting at the required point

Question	Answer	Marks	Part Marks	Guidance
1	circumference	1		Accept any clear indication.
2	78	1		
3	A	1		Accept any clear indication.
4	d and e (alternate) a and b (corresponding) b and c (vertically opposite)	2	Award 1 mark for two correct pairs of angles.	Accept in either order within each pair. For alternate, also accept a and c .
5	1 64 121	3	Award 1 mark for each correct number or if 0 scored, for 2^6 and 11^2 seen.	
6	$26(\text{cm}^2)$	1		
7	13	1		
8	96 tiles	1		Accept any clear indication.
9	Mike gets $\frac{2}{7}$ of the money.	1		Accept any clear indication.
10	$103(^{\circ})$	1		
11(a)	8	1		
11(b)	$6n - 4$ or equivalent	2	Award 1 mark for $6n + c$ or $kn - 4$ where $k \neq 0$	Accept unsimplified for 2 marks, e.g. $2 + 6(n - 1)$
12	parallelograms	1		Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance
13(a)	Ticks Oliver is not correct and A correct explanation, e.g. He has not used the correct order of operations.	1		Accept any equivalent explanation, e.g. He has not used BIDMAS. Accept correct working/answer shown instead of an explanation. Incorrect work/reason (even if with something correct) = 0 marks for the question.
13(b)	$[\pm]10$	2	Award 1 mark for $9 \times 8 + 28$ or better.	Or better, e.g. 100 , $\sqrt{9 \times 8 + 28}$, $\sqrt{72 + 28}$
14	A correct bearing of 285° drawn. 	1		Tolerance $\pm 2^\circ$. Accept without 285° indicated.

Question	Answer	Marks	Part Marks	Guidance
15	$1\frac{5}{6}$ correct answer only	3	Award 2 marks for an equivalent fraction, not a mixed number or in simplest form, e.g. $\frac{11}{6}$, $\frac{22}{12}$, $1\frac{10}{12}$ or Award 1 mark for $[5]\frac{7}{12} - [3]\frac{9}{12}$ or $4\frac{19}{12} - 3\frac{9}{12}$ or $\frac{67}{12} - \frac{45}{12}$	For 1 mark, accept equivalent fractions with a common denominator, e.g. $[5]\frac{28}{48} - [3]\frac{36}{48}$ For 1 mark, accept other equivalent methods, e.g. $2\frac{7}{12} - \frac{2}{12}$, $2 - \frac{2}{12}$
16	$A = (-2, -1)$ $B = (4, 7)$	2	Award 1 mark for each correct answer.	
17	39.3	2	Award 1 mark for 1 kilometre = $\frac{5}{8}$ mile or 1 mile = 1.6 kilometre or equivalent seen or implied.	Implied by answer 35
18	28 38	2	Award 1 mark for each correct answer.	

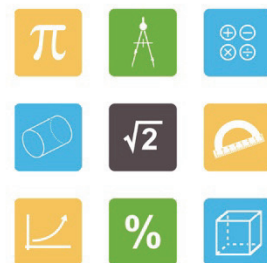
Question	Answer	Marks	Part Marks	Guidance																				
19	<table><tr><td></td><td>Must be true</td><td>Could be true</td><td>Must be false</td></tr><tr><td>divisible by 4</td><td>✓</td><td></td><td></td></tr><tr><td>divisible by 5</td><td></td><td></td><td>✓</td></tr><tr><td>divisible by 8</td><td></td><td></td><td>✓</td></tr><tr><td>divisible by 9</td><td></td><td>✓</td><td></td></tr></table>		Must be true	Could be true	Must be false	divisible by 4	✓			divisible by 5			✓	divisible by 8			✓	divisible by 9		✓		2	Award 1 mark for two or three correct answers.	Accept any clear indication.
	Must be true	Could be true	Must be false																					
divisible by 4	✓																							
divisible by 5			✓																					
divisible by 8			✓																					
divisible by 9		✓																						
20	<table><tr><td><i>x</i></td><td>(−2)</td><td>−1</td><td>(2)</td></tr><tr><td><i>y</i></td><td>−5</td><td>(−3)</td><td>3</td></tr></table> and Correct ruled graph extending at least between (−2, −5) and (2, 3).	<i>x</i>	(−2)	−1	(2)	<i>y</i>	−5	(−3)	3	3	Award 2 marks for correctly completed table. or Award 1 mark for two correct values in the table or correctly plotting <i>their</i> three coordinates.													
<i>x</i>	(−2)	−1	(2)																					
<i>y</i>	−5	(−3)	3																					
21	A correct demonstration, e.g. • Lists factors of 36 and 60 to 12 • Common prime factors are $2 \times 2 \times 3 \neq 6$ • $36 \div 6 = 12$ and $60 \div 6$ is 10 10 and 12 have a common factor of 2	1		Accept ‘The correct HCF is 12’																				
22	Two angles with a sum of 180°	1		Accept if <i>x</i> is acute and <i>y</i> is obtuse. Do not accept 90° for either angle.																				

Question	Answer	Marks	Part Marks	Guidance
23	$6x(2x + 3 - 5y)$	2	Award 1 mark for partially factorised expression, e.g. $2x(6x + 9 - 15y)$, $3x(4x + 6 - 10y)$, $6(2x^2 + 3x - 5xy)$, $2(6x^2 + 9x - 15xy)$, $3(4x^2 + 6x - 10xy)$, $x(12x + 18 - 30y)$ or for fully factorised expression with two out of three correct terms inside the brackets.	
24	5 and 2 and 2	2	Award 1 mark for two correct powers or for $[48 =] 2 \times 2 \times 2 \times 2 \times 3$ or for $[150 =] 2 \times 3 \times 5 \times 5$ or for $[7200 =] 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5$	Powers must be in the correct order. For 1 mark, accept as e.g. factor trees, repeated division, listing.

Question	Answer	Marks	Part Marks	Guidance																		
25(a)	<table><tr><td>5</td><td>1</td><td>7</td><td></td><td></td><td></td></tr><tr><td>6</td><td>0</td><td>5</td><td>9</td><td></td><td></td></tr><tr><td>7</td><td>1</td><td>2</td><td>6</td><td>8</td><td>8</td></tr></table> Key: 5 1 represents 5.1 (cm)	5	1	7				6	0	5	9			7	1	2	6	8	8	3	Award 2 marks for - fully correct diagram but no/incorrect key - correct key with two fully correct rows - correct key with complete unordered diagram. or Award 1 mark for two fully correct rows - a correct key - complete unordered diagram with no/incorrect key.	Key does not have to use values 5 and 1
5	1	7																				
6	0	5	9																			
7	1	2	6	8	8																	
25(b)	A correct comparative statement of the medians including statistics, e.g. On average the freshwater fish are 7(cm) long which is shorter (than the saltwater fish) or equivalent. and A correct comparative statement of the ranges including statistics, e.g. The saltwater fish vary more in length (than the freshwater fish) as freshwater fish have a range of 2.7(cm) or equivalent.	4	Award 1 mark for On average, freshwater fish are shorter or equivalent and Award 1 mark for Median (of freshwater fish) = 7(cm). and Award 1 mark for The saltwater fish vary more in length or equivalent and Award 1 mark for Range (of freshwater fish) is 2.7(cm).	Or equivalent, e.g. - On average the freshwater fish are 1.4 (cm) shorter. - The median for freshwater fish is 7(cm) which is shorter. and - The range of freshwater fish is 2.7(cm). Their lengths are more consistent. Do not accept comparisons without context, e.g. The saltwater fish have a greater range than the freshwater fish. For the comparative statements only, accept follow through for <i>their</i> median and <i>their</i> range in part (a).																		

Question	Answer	Marks	Part Marks	Guidance
26	$(A =) (7, 4)$ $(B =) (7, 9)$	2	Award 1 mark for each correct pair of coordinates.	

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Mathematics

Stage 8

Paper 2

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Calculator
Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Expand the brackets.

$$5x(x + 4)$$

..... [1]

- 2 Rajiv sells three different sizes of cups of coffee in the ratio

$$\text{large} : \text{medium} : \text{small} = 3 : 9 : 2$$

He sells 238 cups of coffee in total.

Work out how many **small** cups of coffee Rajiv sells.

..... [2]

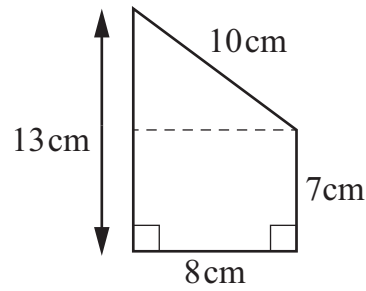
- 3 Here is an inequality.

$$x \leq 7$$

Write down the greatest possible value of x .

$x =$ [1]

- 4 Here is a compound shape made from a triangle and a rectangle.



NOT TO
SCALE

Anastasia wants to work out the area of this shape.

Draw a ring around the side length she does **not** need to use.

7 cm

8 cm

10 cm

13 cm

[1]

- 5 Gabriella says,

‘I start with a number, I square it and the answer is 169’

Find the two possible numbers Gabriella could have started with.

..... and [1]

- 6 A vase contains flowers that are white or red or pink only.
A flower is picked at random from the vase.
The probabilities of picking each colour of flower are shown in the table.

White	Red	Pink
0.15	0.05	0.8

Find the probability of **not** picking a red flower.

..... [1]

- 7 Rearrange the formula to make x the subject.

$$m = \frac{x + y}{2}$$

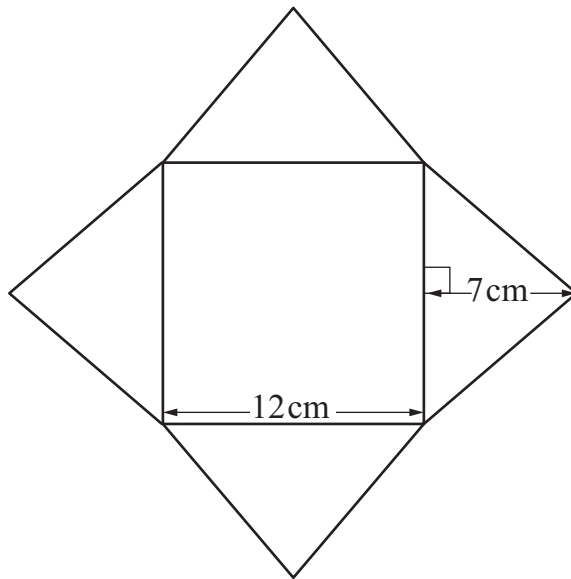
$x =$ [2]

- 8 An expression, in cm, for the width of a rectangle is $x + 3$
The length of the rectangle is twice the width.

Find, in terms of x , an expression for the perimeter of the rectangle.
Give your answer in its simplest form.

..... cm [2]

- 9 Here is the net of a square-based pyramid.



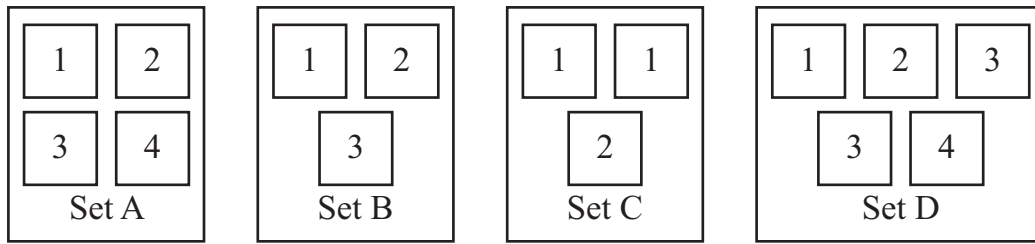
NOT TO
SCALE

The top vertex of the pyramid is directly above the centre of the base.

Calculate the total surface area of the pyramid.

..... cm^2 [2]

10 Yuri has four sets of numbered cards, A, B, C and D.



- (a) Yuri chooses **one** of the sets of cards.
 He picks one card at random from this set and replaces it.
 He does this a total of 600 times from the same set.
 Yuri picks the number 1 a total of 118 times.

Draw a ring around the set Yuri is most likely to have chosen.

Set A

Set B

Set C

Set D

[1]

- (b) Yuri picks one card at random from set A and one card at random from set C.
 He adds the numbers on the two cards together.

- (i) He starts to draw this sample space diagram to show all the possible outcomes.

		Set C		
Set A	+	1		
	1			
	2			
	3	4		
	4			

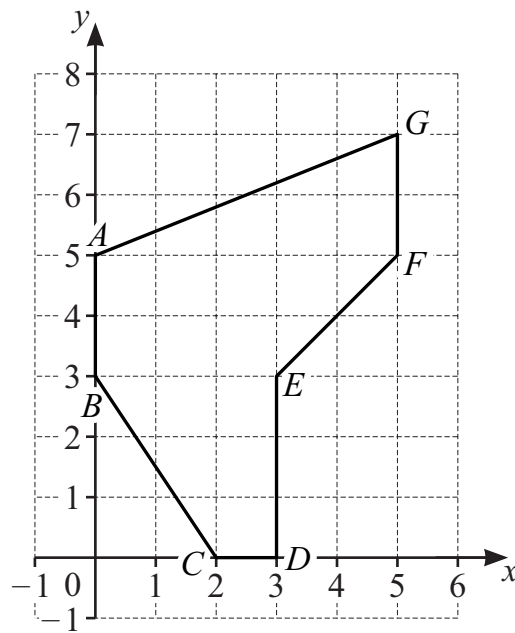
Complete the diagram.

[2]

- (ii) Find the probability Yuri gets a total of 3 when he adds the numbers on his two cards together.

..... [1]

11 The diagram shows shape $ABCDEFGG$ drawn on a grid.



(a) Match each equation to the correct line.

	Line through F and G
$y = 3$	Line through C and D
$x = 0$	Line through D and E
$x = 5$	Line through B and E
$y = 0$	Line through A and F
	Line through A and B

[2]

(b) Write down the equation of the line through E and F .

..... [1]

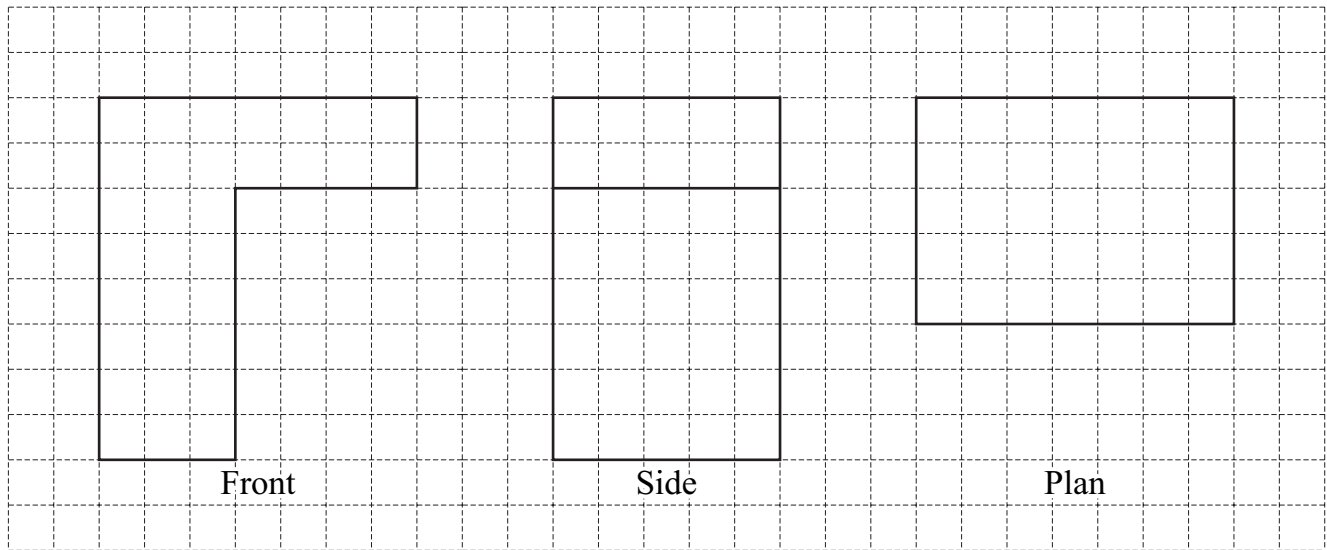
- 12** Mia writes down five whole numbers.
She picks one of the numbers at random.
Outcome E is she picks an even number.
Outcome O is she picks an odd number.

Tick (✓) to show if each of these statements must be true, could be true or must be false.

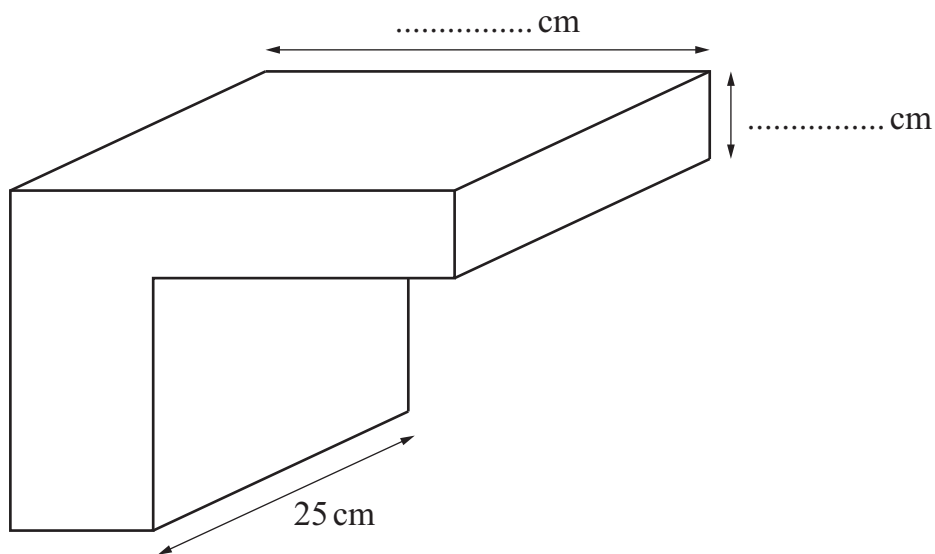
	Must be true	Could be true	Must be false
Outcome E is more likely to happen than outcome O	☐	☐	☐
Outcome E and outcome O can happen at the same time	☐	☐	☐

[1]

13 Ahmed makes these scale drawings of the front view, side view and plan view of a prism.



He then sketches this 3D-drawing of the prism.

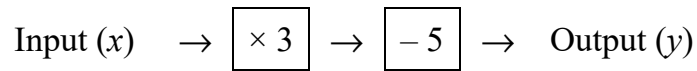


NOT TO
SCALE

Work out the **two** missing lengths in the 3D-drawing.
Write your answers on the drawing.

[2]

14 Here is a function machine.



Complete the table showing inputs and outputs for the function machine.

Input (x)	Output (y)
-10	
	7

[2]

15 Lily wants to investigate the favourite gym equipment of members of her gym.

- (a) There are 700 adults and 50 children who are members of her gym.
Lily decides to sample 30 adults and 30 children.

Give one reason why this may **not** be an appropriate sample to choose.

.....
..... [1]

- (b) Lily decides to interview the members of her sample instead of giving them a questionnaire.

Write down one advantage of this method.

.....
..... [1]

- 16** A box contains green, red and black pens only.

The proportion of the pens that are green is $\frac{5}{12}$

The proportion of the pens that are red is $\frac{1}{3}$

Find the ratio of green to red to black pens.

Give the ratio in its simplest form.

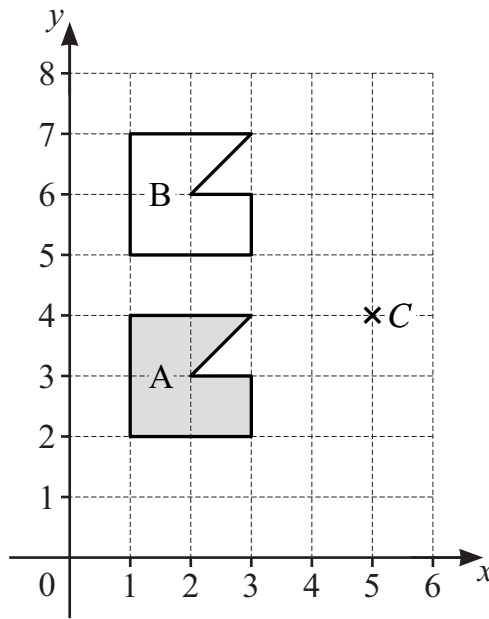
..... : : [2]

- 17** The circumference of a circle is 136.4 cm.

Calculate the radius of the circle.

..... cm [2]

18 The grid shows shape A, shape B and point C.



(a) Write down the vector that translates shape A onto shape B.

$$\begin{pmatrix} \dots\dots\dots \\ \dots\dots\dots \end{pmatrix}$$

[1]

(b) Point C is translated to point D by vector $\begin{pmatrix} -25 \\ 40 \end{pmatrix}$.

Write down the vector that translates point D onto point C.

$$\begin{pmatrix} \dots\dots\dots \\ \dots\dots\dots \end{pmatrix}$$

[1]

(c) Point C is reflected in the line $x = 6$ to give point E.

Find the coordinates of point E.

(..... ,) [1]

19 Complete these sentences with the correct rounded number.

307 021 correct to 2 significant figures is

0.0389647 correct to 3 significant figures is

[2]

20 (a) Complete these sentences about percentage change with the correct numbers.

To increase a value by 4% you multiply by the decimal

To increase a value by% you multiply by 2

[2]

(b) Complete this sentence about absolute change with the correct number.

When 150 is decreased by 300% the absolute change is

[1]

21 Mike writes three equations where

- h represents the number of hours
- d represents the number of days
- m represents the number of months
- y represents the number of years

Tick (✓) to show if each of his equations is always true, sometimes true or false.
One has been done for you.

	Always true	Sometimes true	False
$d = 24h$	☐	☐	☒
$m = 12y$	☐	☐	☐
$d = 31m$	☐	☐	☐

[1]

- 22** Angelique has some red tickets and some yellow tickets.
Each ticket has a single number written on it.
The table shows information about her tickets.

Colour of the ticket	Numbers used on the tickets
red	1 to 27
yellow	1 to 20

Angelique picks a ticket at random.
She writes these sentences about the ticket she picks, each with an incorrect fraction.

The probability that my ticket is red is $\frac{1}{2}$

The probability that my ticket shows an even number is $\frac{1}{2}$

Complete her sentences with the correct fractions.

The probability that my ticket is red is

The probability that my ticket shows an even number is

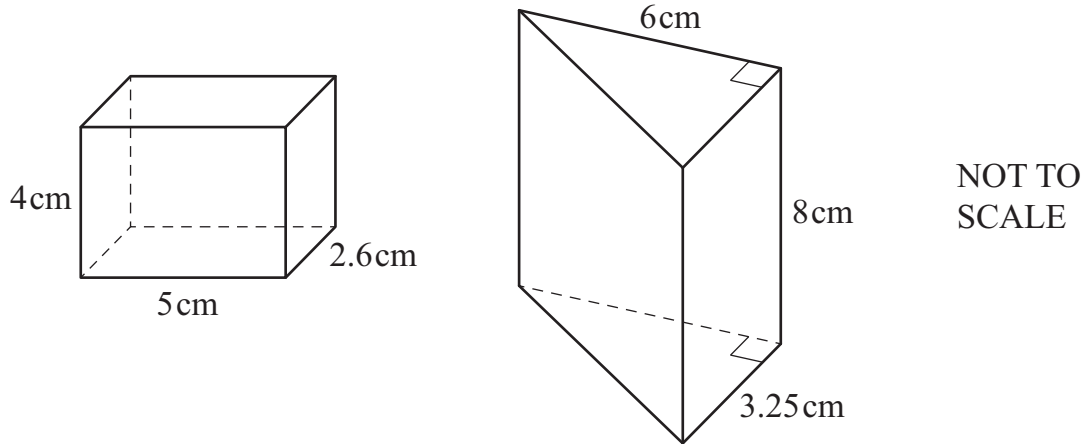
[2]

- 23** Solve.

$$5(4x + 3) = 97 - 8(0.5 - x)$$

$$x = \dots\dots\dots [3]$$

- 24 The diagram shows a cuboid container and a triangular prism container.
The cross-section of the triangular prism is a right-angled triangle.



The cuboid is completely filled with water.
The water is poured from the cuboid into the empty triangular prism.

Calculate the **fraction** of the triangular prism that is filled by the water.
Give your answer in its simplest form.

..... [4]

25 Here is an equation.

$$8 - x + y = 20$$

Choose the correct words and the correct number to complete the sentence.

greater than	less than
--------------	-----------

8	20	12	28
---	----	----	----

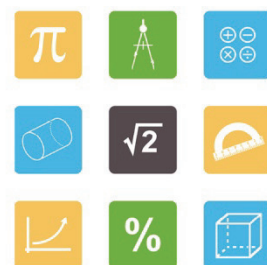
x is y by

[1]

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Mathematics

Stage 8

Paper 2

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Mark scheme annotations and abbreviations**

FT	follow through after error
dep	dependent
oe	or equivalent
cao	correct answer only
isw	ignore subsequent working
soi	seen or implied

Difference in printing

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square in brackets in mark scheme

When Square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0–30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

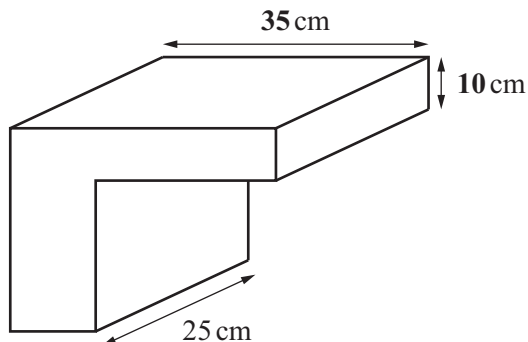
The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	$5x^2 + 20x$	1		
2	34	2	Award 1 mark for $238 \div (3 + 9 + 2)$ or better.	Or better, e.g. $238 \div 14, 17,$ $\frac{2}{3+9+2} \times 238$
3	$(x =) 7$	1		
4	10 cm	1		Accept any clear indication.
5	13 and -13	1		Accept ± 13 with no incorrect answers.
6	0.95	1		Accept equivalent fractions and percentage.
7	$(x =) 2m - y$	2	Award 1 mark for correct first step, e.g. $2m = x + y$ or $m - \frac{y}{2} = \frac{x}{2}$	
8	$6x + 18$ or $6(x + 3)$ (cm)	2	Award 1 mark for a correct expression not in its simplest form, e.g. $x + 3 + x + 3 + 2(x + 3) + 2(x + 3)$ or final answer $6x + k$	
9	$312(\text{cm}^2)$	2	Award 1 mark for $\frac{1}{2} \times 12 \times 7 [\times 4]$ or equivalent or 12×12 or equivalent	Implied by 42, 168 or 144 seen.

Question	Answer	Marks	Part Marks	Guidance																											
10(a)	Set D	1		Accept any clear indication.																											
10(b)(i)	<table border="1"> <tr><td></td><td></td><td colspan="3" style="text-align: center;">(Set C)</td></tr> <tr><td></td><td style="text-align: center;">+</td><td style="text-align: center;">(1)</td><td style="text-align: center;">1</td><td style="text-align: center;">2</td></tr> <tr><td rowspan="4" style="text-align: center; vertical-align: middle;">(Set A)</td><td style="text-align: center;">(1)</td><td style="text-align: center;">2</td><td style="text-align: center;">2</td><td style="text-align: center;">3</td></tr> <tr><td style="text-align: center;">(2)</td><td style="text-align: center;">3</td><td style="text-align: center;">3</td><td style="text-align: center;">4</td></tr> <tr><td style="text-align: center;">(3)</td><td style="text-align: center;">(4)</td><td style="text-align: center;">4</td><td style="text-align: center;">5</td></tr> <tr><td style="text-align: center;">(4)</td><td style="text-align: center;">5</td><td style="text-align: center;">5</td><td style="text-align: center;">6</td></tr> </table>			(Set C)				+	(1)	1	2	(Set A)	(1)	2	2	3	(2)	3	3	4	(3)	(4)	4	5	(4)	5	5	6	2	Award 1 mark for eight or more correct entries.	For 1 or 2 marks, accept the columns in either order.
		(Set C)																													
	+	(1)	1	2																											
(Set A)	(1)	2	2	3																											
	(2)	3	3	4																											
	(3)	(4)	4	5																											
	(4)	5	5	6																											
10(b)(ii)	$\frac{3}{12}$ or equivalent	1	Follow through from <i>their</i> completed table in part (b)(i).	Accept equivalent percentage, decimal or fractions.																											
11(a)		2	Award 1 mark for two or three correct lines.																												
11(b)	$y = x$	1																													

Question	Answer			Marks	Part Marks	Guidance
12	Must be true Outcome E is more likely to happen than outcome O	Could be true ✓	Must be false ✓	1		Both answers correct for the mark. Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance						
13		2	Award 1 mark for recognition of scale 1 square = 5 cm implied by one correct answer.							
14	<table border="1"><thead><tr><th>Input (x)</th><th>Output (y)</th></tr></thead><tbody><tr><td>(-10)</td><td>-35</td></tr><tr><td>4</td><td>(7)</td></tr></tbody></table>	Input (x)	Output (y)	(-10)	-35	4	(7)	2	Award 1 mark for each correct answer.	
Input (x)	Output (y)									
(-10)	-35									
4	(7)									
15(a)	A correct reason, e.g. The sample does not contain the same proportion of adults and children as the population.	1		Accept equivalent statements implying the proportions are wrong, e.g. There are too many children in the sample (as there are only 50 in the population). Do not accept ‘Not representative of the population’ alone.						
15(b)	A correct reason, e.g. • She can explain any questions that are not understood. • She is more likely to get a response.	1		Accept any correct reason in favour of an interview over a questionnaire.						

Question	Answer	Marks	Part Marks	Guidance
16	5 : 4 : 3	2	Award 1 mark for an equivalent ratio not in its simplest form, e.g. $\frac{5}{12} : \frac{1}{3} : \frac{1}{4}$ or $\frac{4}{12}$ and $\frac{3}{12}$ seen.	
17	21.7 (cm) or 21.70 to 21.72 (cm)	2	Award 1 mark for $136.4 \div \pi$ or better.	Accept values of π between 3.14 and $\frac{22}{7}$ Accept 22 for 2 marks with correct working or more accurate answer seen. 1 mark implied by 43 or 43.4... or 22
18(a)	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$	1		
18(b)	$\begin{pmatrix} 25 \\ -40 \end{pmatrix}$	1		
18(c)	(7, 4)	1		
19	310 000 correct answer only 0.0390 correct answer only	2	Award 1 mark for each correct answer.	Do not accept 0.039
20(a)	1.04 100	2	Award 1 mark for each correct answer.	
20(b)	-450	1		Do not accept 450

Question	Answer	Marks	Part Marks	Guidance
21	Always true Sometimes true False $d = 24h$ ☐ ☐ ☐ {✓} $m = 12y$ ☐ ✓ ☐ ☐ $d = 31m$ ☐ ☐ ✓ ☐	1		Both answers correct for the mark. Accept any clear indication.
22	(red) $\frac{27}{47}$ correct answer only (even) $\frac{23}{47}$ correct answer only	2	Award 1 mark for one correct or for both given as decimals correct to 2dp or better, e.g. 0.57446... and 0.48936...	
23	$(x =) 6.5$	3	Award 1 mark for $20x + 15 = 97 - 4 + 8x$ with at most one term incorrect or better. and Award 1 mark for correctly collecting <i>their</i> letters on one side and <i>their</i> numbers on the other, i.e. $20x - 8x = 97 - 4 - 15$	Accept $\frac{13}{2}$ or equivalent for 3 marks. Or better, e.g. $12x = 78$

Question	Answer	Marks	Part Marks	Guidance
24	$\frac{2}{3}$ correct answer only	4	Award 3 marks for correct fraction not in its simplest form, e.g. $\frac{52}{78}$, $5\frac{1}{3}$ or Award 2 marks for $\frac{1}{2} \times 6 \times 3.25 \times 8$ or 78 or $4 \times 5 \times 2.6 \div (\frac{1}{2} \times 6 \times 3.25)$ or $5\frac{1}{3}$ or Award 1 mark for any of these seen • $4 \times 5 \times 2.6$ or 52 • $\frac{1}{2} \times 6 \times 3.25$ or 9.75 • $6 \times 3.25 \times 8$ or 156	
25	less than and 12	1		Both answers in the correct order for the mark.

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